

Scientists are researching climate engineering to reduce global warming. A major technique related this is the cirrus cloud rarefaction technique. Cirrus clouds are icy crystals of water vapor, which prevent heat reflected from the ground from escaping into outer space. By spraying bismuth triiodide or silver iodide on these clouds they, become sparse. As a result of their rarefaction, their heat absorption capacity is reduced, due to which the terrestrial radiation will go comparatively more into space and thus global warming will be reduced. Similarly, under another proposed geoengineering technique, spraying of sulfate aerosols in the stratosphere would create a layer, that reflects some amount of solar radiation before it reaches the Earth's troposphere, thereby reducing radiation on Earth and thus reducing global warming.

**7. Why are dewdrops not formed on a cloudy night?**

- (a) Clouds absorb the radiation released from the Earth's surface.
- (b) Clouds reflect back the Earth's radiation.
- (c) The Earth's surface would have low temperature on cloudy nights.
- (d) Clouds deflect the blowing wind to ground level.

**I.A.S. (Pre) 2019**

**Ans. (b)**

The most suitable conditions for dew formation are clear sky, calm air, high relative humidity and cool long nights. With higher temperatures during the day, the water holding capacity of the air increases, due to which the amount of water vapor (humidity) in the air increases. On the contrary, when the temperature becomes very low at night, the water-holding capacity of the air decreases due to which more water vapor from the saturated air condenses in the form of drops on the cold surface, which is called 'dew'. But when clouds cover the night sky, they reflect back Earth's radiation, keeping the air and surface warm. Due to this the diurnal temperature difference is less and dew does not form.

## The Winds

\*The movement of air as a result of the difference in horizontal air pressure is known as **wind**. Wind always flow from a high- pressure area to a low-pressure area. If the earth were still and flat the winds would blow from a high-pressure area directly to a low-pressure area along the isobaric lines

at an angle of 90°. However, the Earth is geoid in shape and rotates on its own axis. Hence winds under the influence of dynamic forces keep shifting their trajectory.

**Types of Winds** – There are basically three types of wind-

**Planetary Winds** – Winds blowing constantly in a particular direction throughout the year are called planetary winds. The main planetary winds are – **trade winds, westerlies** and **polar winds**. \*Trade winds are winds blowing from sub-tropical high-pressure belt toward equatorial low-pressure belts in both the hemispheres. \***Westerlies**- They blow from sub-tropical high-pressure belts (30°-35°) toward sub-polar low-pressure belts (60°-65°) in both the hemisphere. Since there is more landmass in Northern Hemisphere compared to Southern Hemisphere the westerlies in the former case are more complex, and they are less influential in summers as compared to winter. \*Due to the presence of less landmass in the Southern Hemisphere the westerlies travel at such a high speed causes them to become stormy. It is because of this high intensity of the westerlies in the southern hemisphere that around 40° latitudes they are known as **Roaring Forties**, around 50° latitude they are called **Furious Fifties** and around 60° latitude they are called **Shrieking Sixties**. \*Wellington (New Zealand) is the only capital in the world that lies in the path of Roaring Forties. \***Polar winds** blow from polar high-pressure belt to **sub-polar low-pressure belt**.

**Periodic winds** – Winds are those reverse their direction as per season or time. The important periodic winds are – monsoonal winds, land and sea breezes, breezes of mountains and valleys.

**Monsoonal winds** are those that completely reverse their direction according to seasons. During the six months of summer they blow from sea to land and during the six month of winter they blow from land to sea.

\***Land and Sea Breezes** only cover 20 to 30 km of the area including the coastal area. During daytime the land heats up faster than the nearby sea as a result low pressure is created over land and thus cool and moist winds blow from sea to land. This wind is called a **sea breeze**. There is a complete reversal in the direction of the wind during the night. As a result, winds blow from land to sea. This wind is called **Land breeze**.

• **Mountain and Valley Breeze** - During daytime the air heats up and rises along the slope of the mountain. This is known as **Valley Breeze** (Anabatic flow). During the night the air gets cooler and heavier as a result it starts descending along the slope of the mountain. This is known as **Mountain Breeze** (Katabatic flow).

**(3) Local Winds** – These winds are generated as a result of

regional differences in temperature or air pressure. Hence, they affect a very small area.

\*The Mediterranean Sea is located between 30° to 40° North Latitudes. Hence, the sea comes under the influence of the sub-tropical high-pressure belt. This belt is mobile in nature. During summers, the trade winds blow in these latitudes and in winter these latitudes come under the influence of westerlies. This means that the westerlies blow towards North east from this region and Trade Winds blow towards south west from this region.

1. With reference to "Coriolis force", which of the following statements is/are correct?

1. It increases with increase in wind velocity.
2. It is maximum at the poles and is absent at the equator.

Select the answer using the code given below:

- (a) 1 only                      (b) 2 only  
(c) Both 1 and 2              (d) Neither 1 nor 2

I.A.S. (Pre) 2024

Ans. (c)

The rotation of the Earth about its axis affects the direction of the wind. This force is called the Coriolis force after the French Mathematician who described it in 1844. It deflects the wind to the right in the northern hemisphere and to the left in the southern hemisphere. The deflection is greater when the wind velocity is high, that means Coriolis force increases with an increase in wind velocity. The Coriolis force is directly proportional to the angle of latitude. It is maximum at the poles and is absent at the equator.

The Coriolis force acts perpendicular to the pressure gradient force. The pressure gradient force is perpendicular to an isobar. The higher the pressure gradient force, the greater the velocity of the wind and the larger the deflection in the direction of wind. As a result of these two forces operating perpendicular to each other, in low-pressure areas the wind blows around it. At the equator, the Coriolis force is zero and the wind blows perpendicular to the isobars. The low pressure gets filled instead of getting intensified. That is the reason why tropical cyclones are not formed near the equator. Hence, both statements 1 and 2 are correct.

2. Consider the following statements :

1. In the tropical zone, the western sections of the oceans are warmer than the eastern sections owing to the influence of trade winds.
2. In the temperate zone, westerlies make the eastern sections of oceans warmer than the western sections.

Which of the statements given above is/are correct?

- (a) 1 only                      (b) 2 only  
(c) Both 1 and 2              (d) Neither 1 nor 2

I.A.S. (Pre) 2021

Ans. (c)

The direction of trade winds in the tropics is from east to west. Due to this, when the warm terrestrial wind reaches the ocean located to the west of the landmass, it raises the temperature of those oceans. In the temperate zone, the westerly wind blows from west to east, causing the eastern sections of the oceans to become warmer. So both the statements are correct.

3. Assertion (A) : Wind patterns are clockwise in the northern hemisphere and anti-clockwise in the southern hemisphere.

Reason (R) : The directions of wind patterns in the northern and the southern hemisphere are governed by the Coriolis effect.

Code :

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A)  
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A)  
(c) (A) is true, but (R) is false  
(d) (A) is false, but (R) is true

I.A.S. (Pre) 2005

Ans. (a)

The horizontal winds near the Earth surface respond to the combined effect of three forces – the pressure gradient force, frictional force and the Coriolis force. The rotation of the Earth on its axis affects the direction of the wind. This force is called the Coriolis force after the French physicist G.G. Coriolis who described this force. It deflects the wind to the right(clockwise) direction in the Northern Hemisphere and to the left(anticlockwise) in the Southern Hemisphere. Hence, both assertion (A) and reason(R) are true and reason (R) is the correct explanation of assertion (A).

4. Assertion (A) : Wind patterns are clockwise in southern hemisphere and anti-clockwise in northern hemisphere.

Reason (R) : The directions of wind patterns in the northern and the southern hemispheres are governed by coriolis effect.

**Code :**

- (a) Both (A) and (R) are individually true, and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are individually true, but (R) is not the correct explanation of (A).
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but (R) is true.

**M.P.P.C.S. (Pre) 2008**

**Ans. (d)**

See the explanation of the above question.

**5. What caused wind to deflect towards left in Southern Hemisphere?**

- (a) Magnetic Field
- (b) Coriolis force
- (c) Temperature
- (d) Pressure

**U.P.P.C.S. (Pre) 2023**

**Ans. (b)**

Coriolis force is the force that is responsible for the deflection of the winds in the Northern and Southern Hemispheres.

It is also known as the impact of Ferrel's law.

According to Ferrel's law, any moving fluid turns right in the Northern Hemisphere while in the Southern Hemisphere it turns left.

This Coriolis force that causes the deflection in both of these hemispheres is generated due to the movement of the Earth on its axis, i.e. rotation.

**6. Westerlies in the southern hemisphere are stronger and persistent than in the northern hemisphere because ?**

1. Southern hemisphere has less landmass as compared to northern hemisphere
2. Coriolis force is higher in the southern hemisphere as compared to the northern hemisphere.

**Which of the statements given above is/are correct?**

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

**I.A.S. (Pre) 2011**

**Ans. (a)**

The westerlies are prevailing winds from the west towards the east between subtropical high-pressure zone ( $30^{\circ}$ – $35^{\circ}$ ) and subpolar low pressure ( $60^{\circ}$ – $65^{\circ}$ ). These winds are more complicated in the northern hemisphere due to the larger expanse of land. They are less active during summer and more active in winter. As they pass over the sea, they carry moisture and cause rain. Due to the lesser expanse of land in the southern hemisphere, these winds become extremely

powerful and turn into a storm. Due to its intensity in the southern hemisphere, it is called 'Roaring Forties' at  $40^{\circ}$  latitude, 'Furious Fifties' at  $50^{\circ}$  latitude and 'Shrieking Sixties' at  $60^{\circ}$  latitude.

**7. When the Coriolis effect is counter balanced by the pressure gradient force the resulting winds is termed as :**

- (a) Polar winds
- (b) Prevailing winds
- (c) Geostrophic winds
- (d) Trade

**Uttarakhand P.C.S. (Pre) 2024**

**Ans. (c)**

The Coriolis force is an apparent force that arises due to Earth's rotation. This force acts to the right of the direction of air motion in the Northern Hemisphere and to the left in the Southern Hemisphere. When the Coriolis force and the horizontal pressure gradient force balance each other, the result is geostrophic winds. These winds flow parallel to the isobars and are typically found at altitudes above the ground where friction is minimal. Geostrophic winds are determined by the distribution of density and pressure at a given altitude.

**8. What are roaring forties, furious fifties and shrieking sixties?**

- (a) Oceanic storms
- (b) Westerlies in Southern hemisphere
- (c) Westerlies in Northern hemisphere
- (d) Pacific Ocean currents

**M.P.P.C.S. (Pre) 2015**

**Ans. (b)**

See the explanation of the above question.

**9. In which hemisphere, roaring forties, furious fifties and shrieking sixties are blowing?**

- (a) Northern Hemisphere
- (b) Southern Hemisphere
- (c) Eastern Hemisphere
- (d) Western Hemisphere
- (e) None of the above / More than one of the above

**67<sup>th</sup> B.P.S.C. (Pre) (Re. Exam) 2022**

**Ans. (b)**

Roaring forties, furious fifties and shrieking sixties winds flow in the Southern Hemisphere of the Earth. They were given these names on the basis of the intensity of the Westerly winds in the Southern Hemisphere.

**10. Consider the following statements about the 'Roaring Forties':**

1. They blow uninterrupted in the northern and southern hemispheres
2. They blow with great strength and constancy

3. Their direction is generally from north - west to east in the southern hemisphere
4. Overcast skies, rain and raw weather are generally associated with them

Which of these statements are correct?

- (a) 1, 2 and 3
- (b) 2, 3 and 4
- (c) 1, 3 and 4
- (d) 1, 2 and 4

I.A.S. (Pre) 2000

Ans. (b)

The **Roaring Forties** are strong westerly winds that blow in the Southern Hemisphere, generally near the latitudes of 40°. Roaring Forties are stormy in nature. The strong north-west to east air currents are caused by the combination of air being displaced from the Equator towards the South Pole and the Earth's rotation. The Roaring Forties bring overcast skies, rain, and cold temperatures. Therefore, statements (2),(3) and (4) are correct.

11. In which of the following regions long forties winds blow:

- (a) North Sea
- (b) Indian Ocean
- (c) Caribbean Sea
- (d) Coral Sea

R.A.S./R.T.S. (Pre) 1993

Ans. (b)

See the explanation of the above question.

12. The term "Roaring Forties" refers to:

- (a) Strong ocean winds between 40-60 degree south
- (b) Strong tidal waves between 40-60 degree south
- (c) The sailors heard a loud stormy voice between 40-60 degree south
- (d) Loud voices of volcanic eruptions between 40-60 degree south in the sea

R.A.S./R.T.S. (Pre) 1993

Ans. (a)

See the explanation of the above question.

13. What causes the wind to deflect toward the left in the southern hemisphere?

- (a) Temperature
- (b) Magnetic field
- (c) Rotation of the earth
- (d) Pressure

I.A.S. (Pre) 2010

Ans. (c)

Deflection of winds to the right in the northern hemisphere and to the left in the southern hemisphere is due to the Coriolis force. Coriolis force arises due to the rotation of the earth on its axis.

14. The winds flowing from high-pressure area toward the Mediterranean sea are-

- (a) Westerly
- (b) Trade wind
- (c) Monsoonal wind
- (d) Sea wind

U.P.P.C.S. (Pre) 1992

Ans. (\*)

The Mediterranean Sea is located in the Northern Hemisphere between 30° - 40° latitude. This area is under a subtropical high-pressure dynamic belt. Summer trade winds and westerly(anti-trade) winter winds blow from these latitude. This means Westerly blow from this latitude toward the south-west to north-east and trade winds flow from these latitudes toward north-east to south-west.

15. The seasonal reversal of winds is the typical characteristic of –

- (a) Equatorial climate
- (b) Mediterranean climate
- (c) Monsoon climate
- (d) All of the above climates

I.A.S. (Pre) 2014

Ans. (c)

Monsoons are those seasonal winds whose direction reverses according to the season. For six months during summer, these winds flow from sea to land and during the winter season, these winds flow from land to sea. According to Dr. Rama Shastri, "Monsoonal winds blow on a large scale over an expanded region. Their direction reverses according to the change in season. According to Dr. H.G . Dubey,"Reversal of the wind system is the keynote of monsoonal climate. Hence, option (c) is the correct answer.

16. What is the effect of Chinook winds in the central plains in the USA?

- (a) Temperature of winter rises
- (b) Temperature of summer reduces
- (c) Uniform temperature
- (d) No effect on temperature

Jharkhand P.C.S. (Pre) 2013

Ans. (a)

Chinook means "snow-eater". Chinook winds are most common during winter and early spring descending along the eastern slopes of the Rocky Mountains from Colorado (USA) into British Columbia (Canada) in the north. Chinook is a hot wind that blows in winter and therefore raises the temperature within a short time. The increase in temperature results in the melting of snow, making pasture land available for grazing animals.